

Supervised and unsupervised learning: We will explore these later in the article

Model training and evaluation: A machine learning model is an abstraction of the question you are trying to answer or the outcome you want to predict. Models are trained and evaluated from existing data.

Training data

When you train a model from data, you use a known data set and make adjustments to the model based on the data characteristics to get the most accurate answer. In Azure Machine Learning, a model is built from an algorithm module that processes training data and functional modules, such as a scoring module.

In supervised learning, if you're training a fraud detection model, you use a set of transactions that are labelled as either fraudulent or valid. You split your data set randomly, and use a part to train the model and a part to test or evaluate the model.

Evaluation data

Once you have a trained model, evaluate the model using the remaining test data. You use data you already know the outcomes for, so that you can tell whether your model predicts accurately.

Other common machine learning terms

- **Algorithm:** A self-contained set of rules used to solve problems through data processing, math, or automated reasoning.
- **Anomaly detection:** A model that flags unusual events or values and helps you discover problems—for example, credit card fraud detection looks for unusual purchases.
- **Categorical data:** This is data that is organised by categories and can be divided into groups. For example, a categorical data set for autos could specify the year of manufacture, the make, model, and price.
- **Classification:** This is a model for organising data points into categories based on a data set, for which category groupings are already known.
- **Feature engineering:** This is the process of extracting or selecting features related to a data set in order to enhance it and improve outcomes. For instance, airfare data could be enhanced by days of the week and holidays.
- **Module:** This is a functional part in a Machine Learning Studio model, such as the Enter Data module that enables entering and editing small data sets. An algorithm is also a type of module in Machine Learning Studio.

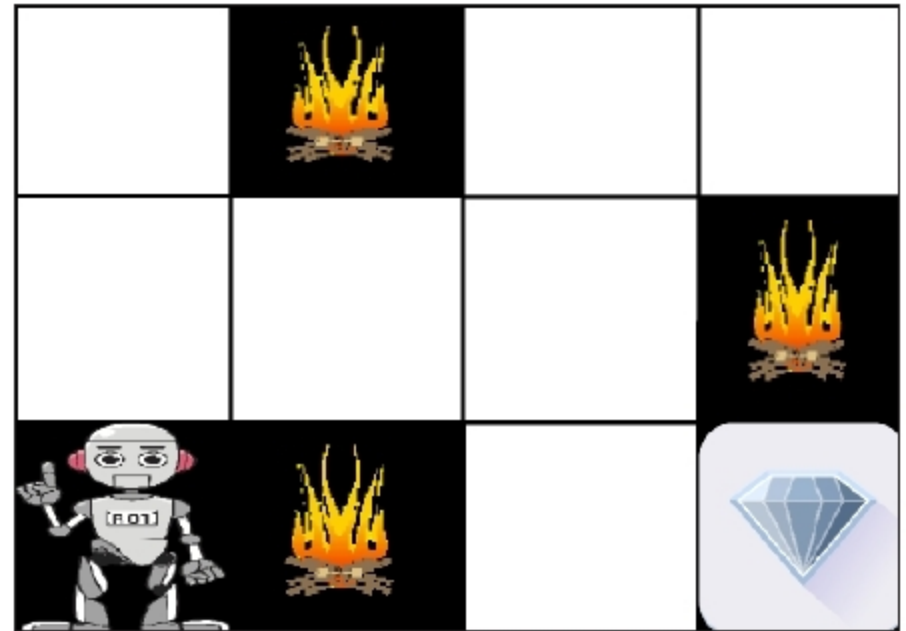


Figure 1: Reinforcement learning example

- **Model:** A supervised learning model is the product of a machine learning experiment comprising training data, an algorithm module, and functional modules, such as a Score Model module.
- **Numerical data:** This is data that has meaning as measurements (continuous data) or counts (discrete data). It is also referred to as quantitative data.
- **Partition:** This is the method by which you divide data into samples (see *Partition* and *Sample* for more information).
- **Prediction:** A prediction is a forecast of a value or values from a machine learning model. You might also see the term 'predicted score'. However, predicted scores are not the final output of a model. An evaluation of the model follows the score.
- **Regression:** This is a model for predicting a value based on independent variables, such as predicting the price of a car based on its year and make.
- **Score:** This is a predicted value generated from a trained classification or regression model, using the Score Model module in Machine Learning Studio. Classification models also return a score for the probability of the predicted value. Once you've generated scores from a model, you can evaluate the model's accuracy using the Evaluate Model module.
- **Sample:** This is a part of a data set intended to be representative of the whole. Samples can be selected randomly or based on specific features of the data set.

What is machine learning?

According to Arthur Samuel, machine learning algorithms enable computers to learn from data, and even improve themselves, without being explicitly programmed.

The basic principle of machine learning is to build algorithms that can receive input data, and use statistical analysis to predict an output while updating outputs as new data becomes available.